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Stress-Sensitive Hollow Proppants for Enhanced Heat Transfer in Closed-Loop Geothermal Systems

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Abstract

Closed-loop geothermal systems (CLGS) face a persistent bottleneck in heat extraction due to the absence of forced convection. Fractured-CLGS was proposed as solution. This study introduces hollow metallic proppants as a new class of heat-transfer media and evaluates their stress-dependent performance using a decoupled discrete element, finite element, and finite volume modeling approach. Results show that copper, aluminum, brass, and bronze consistently outperform sand, with thermal conductivity two to ten times higher across closure stresses up to 144 MPa. Unlike conventional sand or ceramic proppants that maintain static thermal conductivity, metallic proppants exhibit stress-sensitive conductivity. As fractures close, particle rearrangement and contact growth establish thermally conductive bridges that enhance lateral heat transfer. Comparative analysis further reveals trade-offs among thermal conductivity, density, cost, corrosion resistance, and mechanical strength. Copper achieves the highest conductivity but at high density and cost. Aluminum offers a balanced performance with efficient placement. Brass and bronze provide durability in chemically aggressive environments. Stress-sensitive hollow metallic proppants therefore offer a practical pathway to extend the efficiency and longevity of CLGS.

Keywords: Closed-loop geothermal systems, thermally conductive proppants, hollow metallic particles, heat transfer enhancement, proppant pack simulation

Introduction

Closed-loop geothermal systems (CLGS) offer a promising alternative to conventional open-loop geothermal systems (hydrothermal and enhanced geothermal systems). In contrast to open-loop systems that require active fluid circulation within the reservoir, CLGS operate by circulating a working fluid such as water or CO₂ in a sealed wellbore. This means the environmental footprint of such geothermal operation is significantly reduced. The absence of direct contact with the reservoir rock not only reduces the risk of groundwater contamination and scale deposition but also limits the potential for induced seismicity caused by thermal shock since rapid cooling of the formation is avoided (Jahan Bakhsh et al., 2017; Dong et al., 2023). These advantages have attracted significant interest in recent years, especially for geothermal

developments in dry regions or regions with low permeability. Despite these benefits, one of the main technical limitations of CLGS is the absence of direct contact between the working fluid and the geothermal reservoir and reliance on purely conductive heat transfer through the rock mass surrounding the wellbore. Because the working fluid does not interact directly with the formation, convective heat transfer, which is the dominant mode of heat transfer in open-loop systems, is absent. This restricts the overall heat transfer rate and leads to rapid thermal depletion in the near-wellbore region. As heat is extracted, a cooled zone forms around the wellbore and the low thermal conductivity of most geothermal rocks (typically 2–5 W/m·K) prevents heat from being replenished fast enough from the surrounding formation. Recent numerical studies have shown that over a 30-year period, stress redistribution, cooling-induced contraction, and ineffective conductive recharge can lead to local mechanical degradation and steep declines in output temperature (McLean and Espinoza, 2024).

To address this fundamental limitation, researchers explored several strategies to increase effective heat transfer including the use of high-conductivity casings, vacuum-insulated tubing, and fracture-assisted closed-loop systems propped with thermally conductive materials. In the latter approach, hydraulic fractures are induced around the wellbore and filled with thermally conductive materials to promote lateral heat flow toward the wellbore (Liu et al., 2023). Liu et al. (2023) proposed a fractured CLGS that incorporates conductive plugs and fractures filled with metallic-coated proppants. Their simulations showed that this configuration improved net power output by 30.6% through expanding the effective heat exchange surface area. This directed research toward advanced proppant materials and strategies that not only prop the fracture open, but also actively enhance and sustain heat conduction and flow distribution over the project lifetime. Later work by Al Balushi and Dahi Taleghani (2023) introduced metallic coatings such as copper and aluminum on conventional sand and ceramic proppants. In their numerical simulations, conventional sand, ceramic, and bauxite grains are coated with high-conductivity metals such as copper and aluminum. Their results showed that coating sand with a copper layer increased its thermal conductivity from 1.9 W/m·K to 37.48 W/m·K. Their detailed particle-scale simulations confirmed that the thickness of the metallic coating positively correlates with the effective thermal conductivity of the entire proppant pack. This improvement allows the proppant pack to act as a thermal bridge by conducting heat from distant parts of the formation back toward the fracture network and the wellbore.

In this paper, we introduce and evaluate a novel class of hollow, metallic, thermally conductive proppants for application in fractured closed-loop geothermal systems. Hollow metallic proppants offer an additional benefit: high thermal conductivity at significantly lower bulk density. Solid metallic particles may present transportation challenges within the fracture due to their high density (for example, copper has a density of 8.96 g/cm³), requiring higher pumping power and risking uneven proppant distribution within the fracture. Although hollow metallic particles inherently exhibit lower thermal conductivity than solid particles, their reduced bulk density is important for proppant transport and placement. With optimized shell thickness and packing, such proppants can achieve considerable heat transfer performance and serve as lightweight thermal bridges in fractured geothermal systems (Janousch et al., 2016). Metals like brass and bronze have been used in high-temperature, corrosive environments. Manufacturing small hollow spheres of brass or bronze is technically feasible using powder metallurgy technique. The sacrificial templates technique is a proven technique for making hollow metal beads in the millimeter size range. In this process, small polymer or wax spheres are coated with a slurry of metal powder and binder, then heated to burn out the core and sinter the metal shell. Researchers have successfully applied this to steel and titanium, achieving hollow spheres 0.5–3 mm in diameter with wall thickness ~ 0.04–0.25 mm (Andersen et al., 2000). Unlike conventional proppants, the proposed hollow metallic proppants exhibit stress-sensitive thermal conductivity. As closure stress increases, the contact area between proppants increases as well, which improves heat extraction as fracture closure progresses. We utilize finite element method (FEM) to simulate fracture closure and monitor the evolution of the effective thermal conductivity of the proppant pack under representative geothermal conditions. Our work builds upon previous particle- and field-scale studies and aims to quantify the long-

term heat extraction benefits and mechanical resilience of these advanced materials. This approach has the potential to significantly improve the performance and economic viability of CLGS by extending the effective heat exchange surface area and delaying near-wellbore thermal depletion.

Methods

In this study, we follow a decoupled multiphysics approach to test different metallic proppants under representative geothermal reservoir conditions. Our approach combines discrete element method (DEM), finite element method (FEM), and finite volume method (FVM). Fig. 1 shows a summary of the developed workflow.

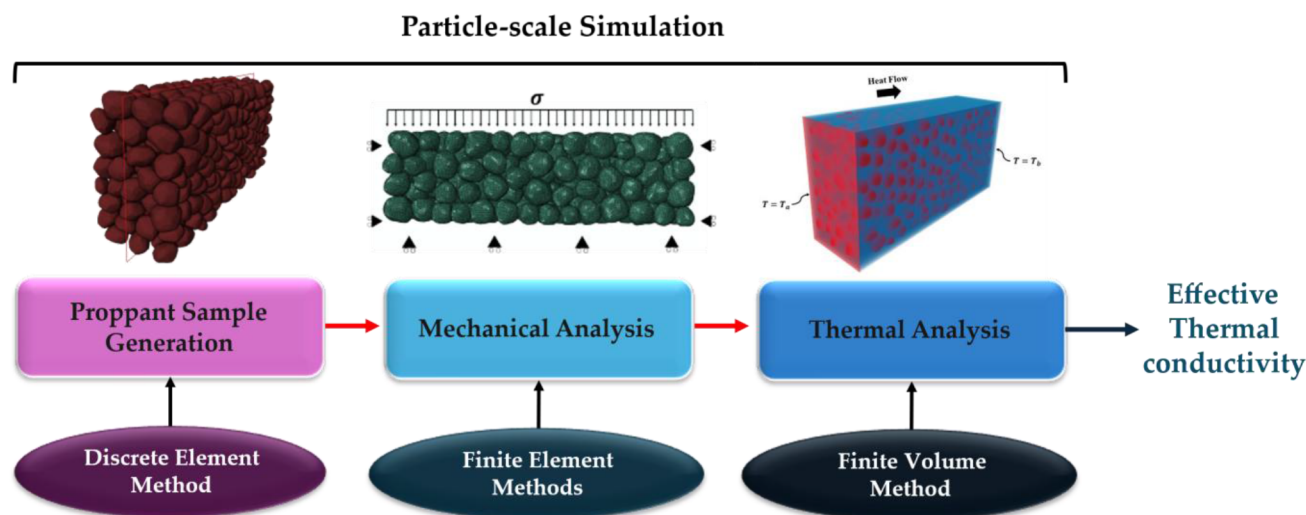


Figure 1—Developed workflow to construct and evaluate the novel hollow proppants through DEM-based particle generation, FEM-based mechanical analysis, and FVM-based heat transfer analysis.

To capture realistic packing morphology, hollow metallic proppants are modeled using DEM (Al Ibrahim et al., 2019). Here, we consider 40/70 mesh with a diameter between 0.212 – 0.42 mm normally distributed. Most numerical studies often consider the grains to be perfect spheres. However, here we consider slightly non-spherical particles to account for any impurities or imperfections that may occur during manufacturing the shells. The process of generating the hollow proppants geometry is a multistep process starting with particle generation using DEM. The pack dimensions, particles size, particle distribution, and friction between particles are specified. The initially generated particles are solid particles similar to the particles shown in Fig. 2a. It is worth noting that proppants are generated under gravitational settling in a confined domain, which represent proppants settling in a fracture. Using image processing, we apply erosion which removes part of the particle near the outer boundary. This reduces the size of the particle. Then, both particles, original and eroded, are subtracted from each other to yield a shell with a thickness equivalent to that removed portion of the particle. Fig. 2d shows a 3D representation of the generated hollow particle.

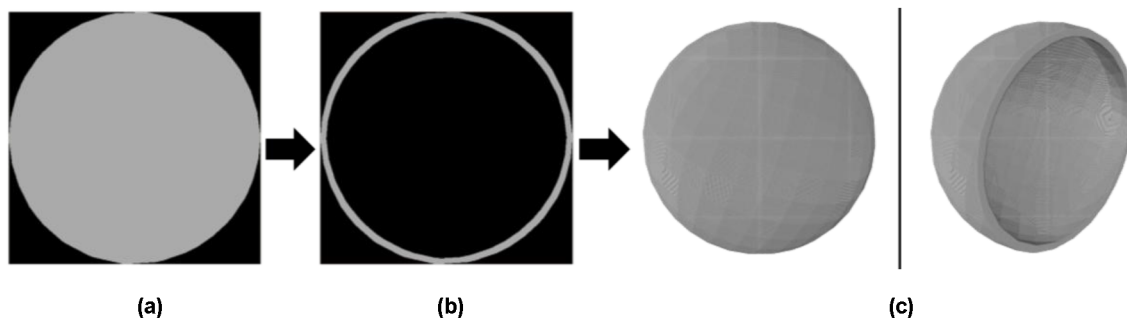


Figure 2—The process of generating hollow particles includes (a) generating solid particles, (b) eroding particles by the desired thickness and subtracting the original and eroded particles. (c) shows a 3D rendering of a coated proppant particle.

The sample generated is then meshed with unstructured mesh to better represent the curved boundaries of the proppant and to result in smoother surfaces, which makes the computations faster and more accurate. The surface mesh is refined using best isotropic triangles algorithm to ensure that the triangular elements have an acceptable aspect ratio of less than 25. Tetra mesh is generated using the enhanced surface mesh and exported for FEM simulations. Here, we test different proppants materials to determine the highest effective thermal conductivity. We consider hollow proppants made of copper, aluminum, brass, and bronze. Before heat transfer is simulated, we subject the proppant sample to in-situ conditions by applying a closure stress of 145 MPa (Fig. 3). We apply the desired effective closure stress by specifying pressure acting on a rigid plate on one side of the geometry. Movement of the rest of the geometry faces is constrained in the normal direction to prevent the granular particles from escaping during the simulation.

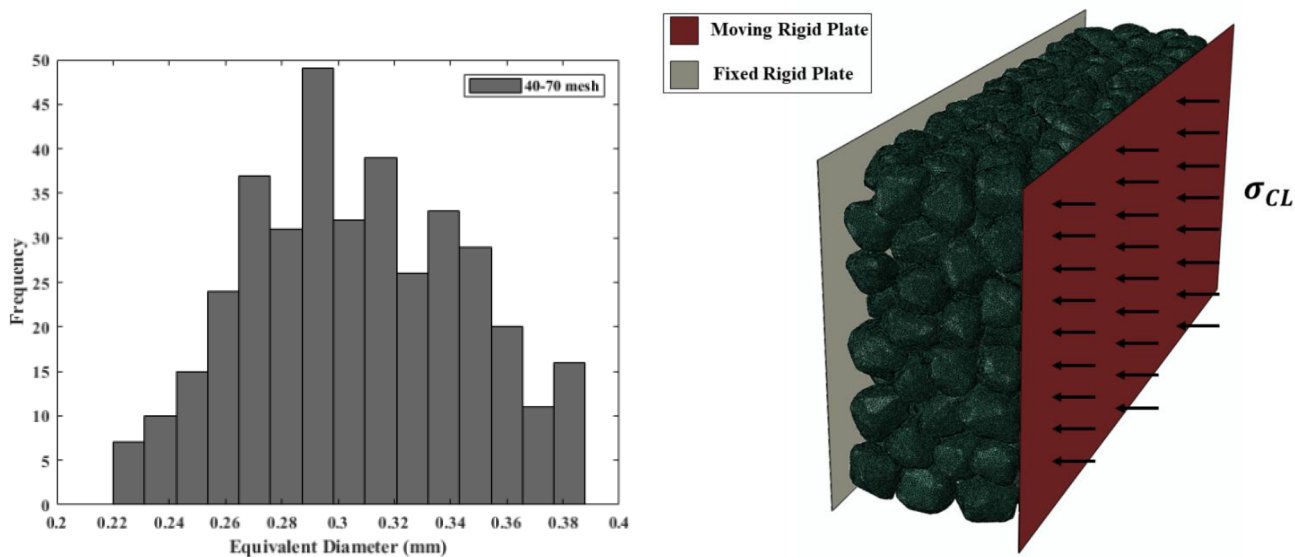


Figure 3—FEM model and boundary conditions used for mechanical simulations.

Once the desired closure stress is reached and the sample has reached mechanical equilibrium, the deformed sample is exported to FVM heat transfer simulations. Using FVM solver, we applied a constant temperature boundary condition of 373.15 K at one end of the sample and a constant temperature of 273.15 K at the opposite end. Other boundaries are assumed to be no-flux boundaries. Effective thermal conductivity is then calculated as

$$k_{t,eff} = \frac{q}{\Delta T / L} \quad (1)$$

where $k_{t,eff}$ is the effective thermal conductivity [$\text{W m}^{-1} \text{K}^{-1}$], q is heat flux per unit area [W m^{-2}], ΔT is the temperature difference between inlet and outlet [K], and L is the length of the sample [m]. Table 1 shows the material properties used for the FEM and FVM simulations.

Table 1—aterial properties used for the FEM and FVM simulations.

Property	Copper	Aluminum	Brass	Bronze
Density [g/cm^3]	8.96	2.7	8.47	8.83
Young's modulus [GPa]	130	70	97	101
Poisson's ratio [-]	0.34	0.33	0.34	0.34
Thermal conductivity [$\text{W/m}\cdot\text{K}$]	398	237	123	64

Results and Discussion

Model validation

To validate the numerical workflow used for effective thermal conductivity calculations, we use a reference case consisting of non-touching spherical particles immersed in a continuous fluid phase. The effective thermal conductivity of this configuration has been analytically calculated by Maxwell (1881) and Meredith & Tobias (1960). Similarly, we use the workflow presented earlier to generate a similar geometry and compute effective thermal conductivity. The same validation approach was previously applied and documented in our earlier study on solid coated proppants. The reader is referred to Al Balushi and Dahi Taleghani (2023) for detailed explanation of the validation case. For completeness and to ensure consistent methodology, we briefly reproduce the validation process here. A cubic domain containing 25 identical spheres (porosity $\sim 62.5\%$) is subjected to a steady-state thermal boundary condition, and the effective conductivity is computed using FVM. The numerical results show good agreement with both analytical solutions, which confirms the reliability of the simulation approach used in this work

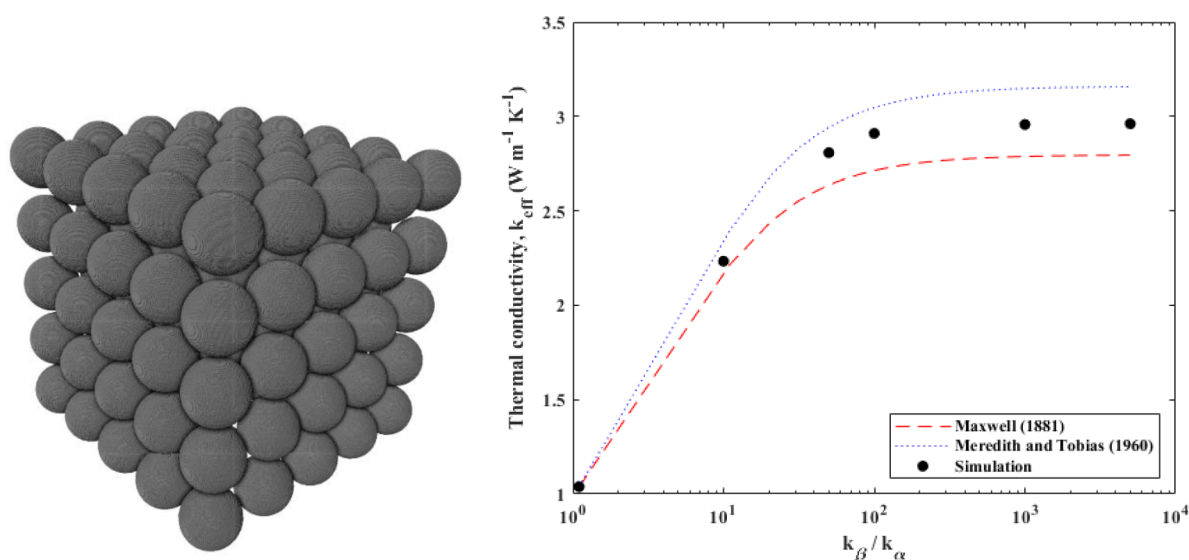


Figure 4—Thermal conductivity computed numerically agrees well with the analytical results (Al Balushi and Dahi Taleghani, 2023).

Thermal Conductivity of Stress-Sensitive Hollow Proppants

Following model validation, we evaluate the thermal performance of four different hollow metallic proppants: copper, aluminum, brass, and bronze under closure stresses representative of geothermal reservoir. Fig 5 shows that all metallic proppants outperform conventional sand across the entire stress range. Initially, at zero closure stress, thermal conductivities of the metallic proppants already exceed that of sand ($1.9 \text{ W/m}\cdot\text{K}$) by a factor of two to three, with copper starting at $5.1 \text{ W/m}\cdot\text{K}$ and aluminum at $4.6 \text{ W/m}\cdot\text{K}$. As fracture closure progresses, conductivities increase steadily, with copper ultimately reaching nearly $18.6 \text{ W/m}\cdot\text{K}$ at 144 MPa , almost ten times higher than sand under the same conditions. Aluminum follows with a maximum of $14.8 \text{ W/m}\cdot\text{K}$, while brass and bronze, though lower in absolute terms (8.4 and $4.7 \text{ W/m}\cdot\text{K}$ respectively), still maintain substantial advantages over conventional sand proppants. This confirms that thermal conductivity is not a fixed property of the hollow proppant pack but evolves dynamically with fracture closure. As the proppant pack compacts, contacts area between particles grow from a point contact to a surface and thermal resistance decreases. in which particle compaction and rearrangement progressively reduce thermal resistance across proppants contact area, forming conductive bridges between particles. In our previous studies on coated sand or ceramic proppants, static conductivity improvements were reported (Liu et al., 2023; Al Balushi & Taleghani, 2023). Our results here show a dynamic response where conductivity continues to improve with stress, implying that thermal efficiency of metallic proppant packs is not fixed upon deployment, but evolves as fractures close.

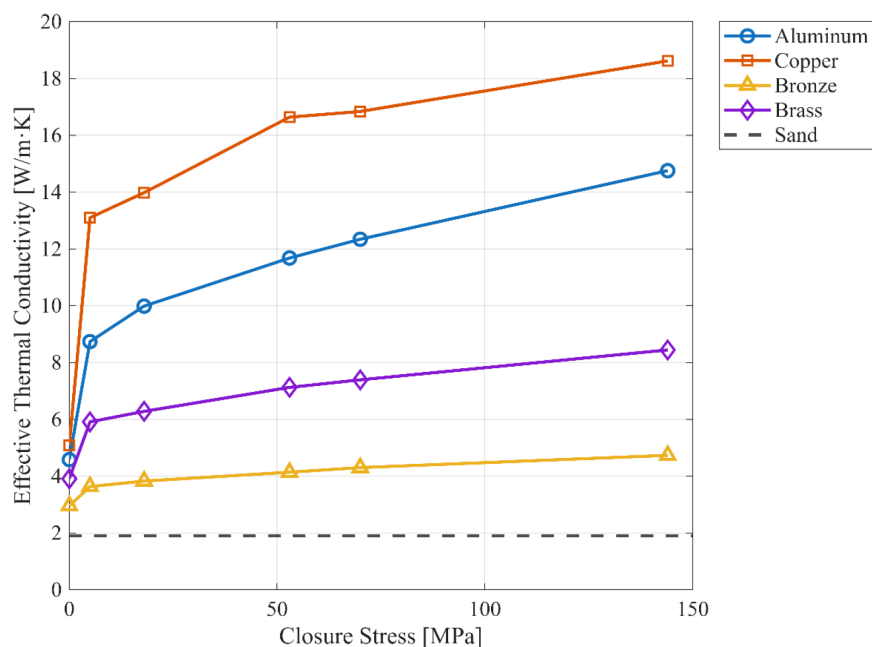


Figure 5—Effective thermal conductivity of hollow copper proppants at different closure stress.

Fig. 6 demonstrates the physical mechanism of dynamic stress-sensitive conductivity. As closure stress increases, proppants deform and rearrange within the fracture. This expands their surface contact areas and establishes conductive bridges across the pack. The metallic shells then act as efficient lateral pathways that channel heat from the hot formation into the cooled near-wellbore zone. This mechanism facilitates heat transfer and explains why thermal conductivity values grow with increasing stress rather than remaining static. The coupling of fracture mechanics and thermal conduction suggests that metallic hollow proppants can be designed not only to withstand stress but to leverage it, turning closure into a process that enhances thermal connectivity rather than degrading it. This stress-activated behavior introduces a fundamentally different perspective on fracture conductivity in geothermal systems: instead of being a property that

diminishes with time and compaction, thermal conductivity can be enhanced through intelligent material design.

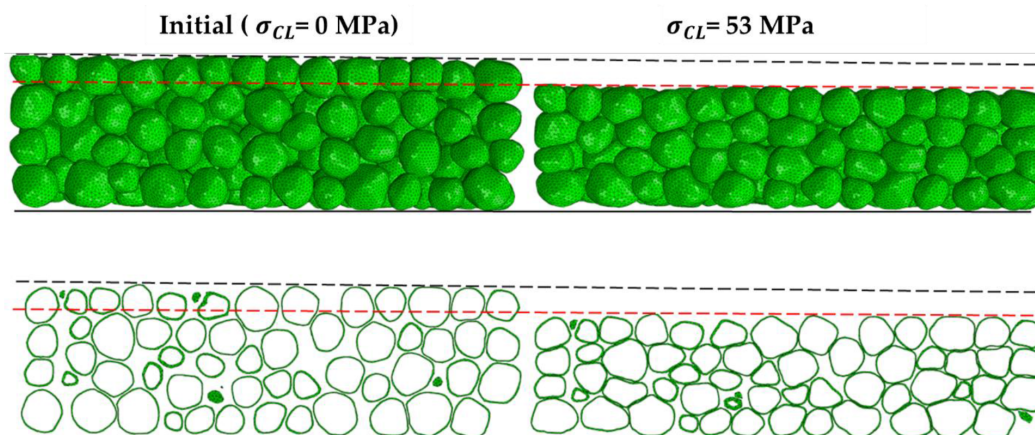


Figure 6—The concept of stress-activated hollow proppants.

To better understand when conductivity gains occur, we analyzed stress sensitivity across three closure zones (0–20, 20–70, and 70–144 MPa) in Fig. 7. The figure tells us when each material delivers its maximum benefit. Copper exhibits a steep increase during the early closure stage (<20 MPa), with an average slope of approximately 0.8 W/m·K per MPa—more than twice that of aluminum and nearly four times that of brass or bronze. This rapid early enhancement suggests that copper proppants could be especially effective in improving near-wellbore conductivity during the initial phases of fracture closure, where heat transfer limitations are often most acute. However, the marginal benefit of copper decreases sharply at higher stresses, plateauing above 70 MPa. Aluminum, by contrast, shows a more uniform gain across both medium (20–70 MPa) and high (70–144 MPa) stress ranges, providing a steadier improvement that may support longer-term heat extraction as fractures fully close. Brass and bronze display smaller absolute sensitivities but maintain consistent slopes across all zones, indicating stable but modest contributions. These differences suggest that material choice should be informed not only by peak thermal conductivity values, but also by the temporal distribution of conductivity gains. Different geothermal reservoirs may require early- versus late-stage enhancement of heat transfer.

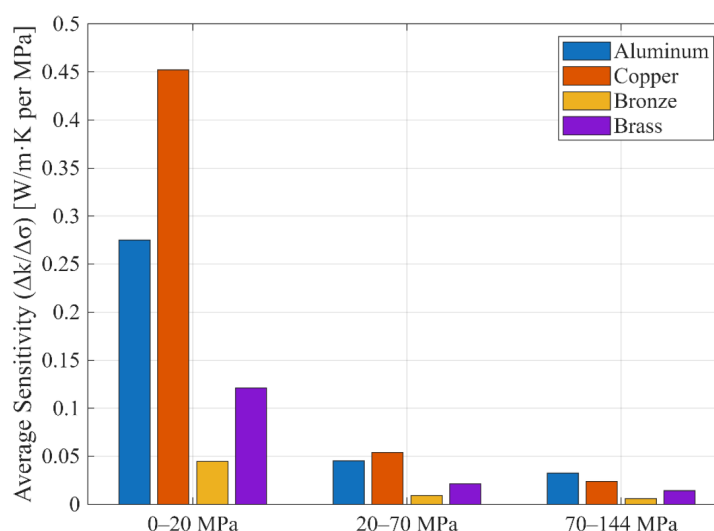


Figure 7—Average stress sensitivity of hollow metallic proppants across three closure-stress zones.

Beyond conductivity alone, real-world deployment requires balancing thermal performance with practical constraints such as density, cost, corrosion resistance, and mechanical strength. Fig. 8 shows a radar chart comparing these factors for the four metals. This analysis shows distinct trade-offs that would not be evident from conductivity data alone. Copper dominates in conductivity and gain but is penalized by high density and cost, which may complicate pumping and transport at scale. Aluminum offers a more balanced profile. It combines moderate conductivity with the lowest density and cost, making it attractive where operational efficiency and transport are critical. Brass and bronze rank lower in conductivity but higher in corrosion resistance and durability. This could be critical in reservoirs containing saline or chemically aggressive fluids. Such an integrated assessment has rarely been presented in prior geothermal studies, which often emphasized thermal properties in isolation. This metric combination suggests copper is best for short-term conductivity-driven applications, aluminum for scalable deployment, and brass/bronze for chemically aggressive reservoirs. This analysis combining mechanical, economic, and chemical considerations alongside thermal properties extends beyond prior studies, which often prioritized conductivity alone, and instead positions proppant selection as a multi-criteria optimization problem.

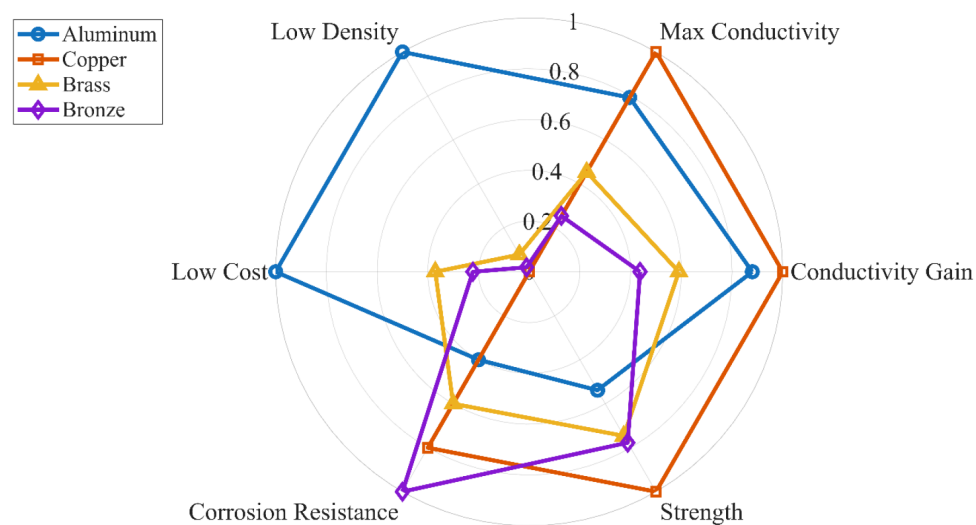


Figure 8—Multi-criteria comparison of different hollow metallic proppants. Metrics include stress-sensitive conductivity gain, maximum conductivity at 144 MPa, density, cost, corrosion resistance, and mechanical strength.

Conclusions and Recommendations

This study introduced hollow metallic proppants as a new class of proppants for enhancing heat transfer in fractured closed-loop geothermal systems. The results showed that these proppants clearly exhibit stress-sensitive thermal conductivity that improves as fractures close, rather than remaining static as in conventional sand. Copper reached nearly 18.6 W/m·K at 144 MPa, almost ten times the conductivity of sand. This shows a promising potential to overcome thermal bottlenecks. Aluminum provided a more balanced profile of moderate conductivity, low density, and low cost that makes it suitable for efficient large-scale deployment. Brass and bronze offered lower conductivity but superior corrosion resistance and durability, which may extend operational life in chemically aggressive environments. These results prove that proppant selection should no longer be guided by thermal conductivity alone, but instead approached as a multi-criteria optimization problem that balances thermal, mechanical, and economic trade-offs. More broadly, this work reframes fracture conductivity in geothermal systems as a dynamic, stress-activated property rather than a fixed parameter, suggesting that closure stress can be leveraged to enhance heat transfer performance. Stress-sensitive metallic proppants therefore provide a practical pathway to tailor subsurface heat-exchange materials to specific reservoir conditions, improving both the efficiency and longevity of closed-loop geothermal operations.

Nomenclature

CLGS	Closed-loop geothermal system
DEM	Discrete Element Method
FEM	Finite Element Method
FVM	Finite Volume Method
k_{eff}	Effective thermal conductivity [$\text{W/m}\cdot\text{K}$]
q	Heat flux per unit area [W m^{-2}]
T	Temperature [K]
L	Length of the sample [m]

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